

Bilkul. Main **exact tumhare code ke according**, line-by-line simple Hinglish mein explain karta hoon, taaki tum directly notes mein copy-paste kar sako.

Butterfly Pattern — Code Explanation

```
int n = 4;
```

- `n = 4` ka matlab butterfly ki **height/size 4** hai.
- Isse pattern ki upper aur lower half decide hoti hai.

1 First Half — Upper Part

```
for (int i = 1; i <= n; i++) {
```

- Ye loop **upper half** print karta hai.
- `i = 1` → first row se start.
- `i <= n` → jab tak `i` 4 hai.
- `i++` → har row ke baad `i` 1 se increase hota hai.

So:

```
i = 1  
i = 2  
i = 3  
i = 4
```

2 First Part — Left Side Stars

```
for (int j = 1; j <= i; j++) {  
    System.out.print("*");  
}
```

- Ye butterfly ke **left side ke stars** print karta hai.
- `j <= i` ki wajah se har row mein `i` number of stars print hote hain.

```
i = 1 → *  
i = 2 → **  
i = 3 → ***  
i = 4 → ****
```

So left side:

```
*  
**
```

```
***
****
```

3 Spaces — Middle Gap

```
int spaces = 2 * (n - i);
```

- Ye middle mein kitne **spaces** chahiye, wo calculate karta hai.
- Formula:

```
spaces = 2 × (n - i)
```

For `n = 4`:

i	Calculation n	Spaces
1	$2 \times (4 - 1)$	6
2	$2 \times (4 - 2)$	4
3	$2 \times (4 - 3)$	2
4	$2 \times (4 - 4)$	0

Isliye upper part mein gap gradually kam hota hai.

4 Spaces Print Karna

```
for (int j = 1; j <= spaces; j++) {
    System.out.print(" ");
}
```

- Ye loop calculated `spaces` ko print karta hai.
- `" "` mein **space character** hai.
- `*` nahi, sirf blank space print hota hai.

Example:

```
*      *
**     **
***    ***
*****
```

5 Second Part — Right Side Stars

```
for (int j = 1; j <= i; j++) {
    System.out.print("*");
}
```

- Ye butterfly ke **right side ke stars** print karta hai.
- Left side ki tarah yahan bhi **i** number of stars print hote hain.

Example:

```
i = 1 → *
i = 2 → **
i = 3 → ***
i = 4 → ****
```

Left + Spaces + Right combine hone par:

```
*      *
**     **
***    ***
*****
```

6 Next Line

```
System.out.println();
```

- Current row complete hone ke baad cursor ko **next line** mein le jata hai.
- Agar `println()` nahi hoga, to saare stars ek hi line mein print honge.

7 Second Half — Lower Part

```
for (int i = n; i >= 1; i--) {
```

- Ye butterfly ka **lower part** print karta hai.
- Yahan **i** reverse direction mein chalta hai.

```
i = 4
i = 3
```

```
i = 2  
i = 1
```

Isliye stars:

```
****  
***  
**  
*
```

8 Lower Part — First Side Stars

```
for (int j = 1; j <= i; j++) {  
    System.out.print("*");  
}
```

- Ye left side ke stars print karta hai.
- `i` decrease ho raha hai, isliye stars bhi decrease hote hain.

```
i = 4 → ****  
i = 3 → ***  
i = 2 → **  
i = 1 → *
```

9 Lower Part — Spaces

```
int spaces = 2 * (n - i);
```

Same formula yahan bhi use hua hai.

```
i = 4 → 0 spaces  
i = 3 → 2 spaces  
i = 2 → 4 spaces  
i = 1 → 6 spaces
```

Isliye lower part mein gap **increase** hota hai.

10 Lower Part — Spaces Print

```
for (int j = 1; j <= spaces; j++) {  
    System.out.print(" ");  
}
```

- Ye calculated number of spaces print karta hai.

- Jaise `i = 2` par:

```
spaces = 2 × (4 - 2)
        = 4
```

Toh 4 spaces print honge.

1 1 Lower Part — Right Side Stars

```
for (int j = 1; j <= i; j++) {
    System.out.print("*");
}
```

- Ye right side ke stars print karta hai.
 - `i` decrease hone ke saath stars bhi decrease hote hain.
-

Complete Logic in Short

Butterfly pattern mein **4 main parts** hain:

```
Left Stars + Spaces + Right Stars
```

Upper Half:

```
i = 1 → *      *
i = 2 → **     **
i = 3 → ***    ***
i = 4 → *      *
```

- Stars **increase**
- Spaces **decrease**

Lower Half:

```
i = 4 → *      *
i = 3 → ***    ***
i = 2 → **     **
i = 1 → *      *
```

- Stars **decrease**
 - Spaces **increase**
-

★ Sabse Important Formula

```
int spaces = 2 * (n - i);
```

Ye butterfly ka **middle gap** control karta hai.

For **n = 4**:

```
i = 1 → 6 spaces  
i = 2 → 4 spaces  
i = 3 → 2 spaces  
i = 4 → 0 spaces
```

Then lower half mein wahi formula reverse effect deta hai:

```
i = 4 → 0 spaces  
i = 3 → 2 spaces  
i = 2 → 4 spaces  
i = 1 → 6 spaces
```

Final Pattern

```
*      *  
**     **  
***    ***  
*****  
*****  
***    ***  
**     **  
*      *
```

Yaad rakhne ka easiest way:

i = stars ki quantity

spaces = middle ka gap

Upper loop → **i** increase

Lower loop → **i** decrease.